Archer Case Study

Safe Autonomy with Runtime Assurance

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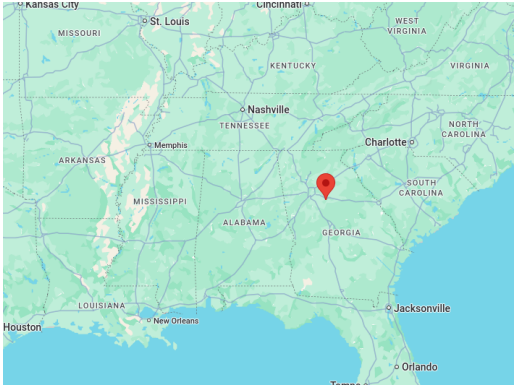
My Start: Covington, GA



Background

B.S. / M.S. / Ph.D. @ Georgia Tech — Aerospace & Robotics 2011–2021

Thesis: *Optimization-Based Approaches to Safety-Critical Control with Applications to Space Systems*

Intern @ ISAE-ENSMA — Quadcopter control 2015

GRA @ Georgia Tech — Safety verification on the Robotarium 2016–2021

Intern @ MIT Lincoln Lab — Neural network verification 2017–2018

Visiting Researcher @ Stanford — Collision-inclusive planning 2018

Intern @ NASA JPL — Spacecraft swarm control 2019

Visiting Student @ KAUST — Safe autonomy, control theory 2020

Intern @ AFRL — Safe spacecraft docking 2020

CEO & Co-Founder @ Pytheia — Camera perception for robots 2021–2024

CTO & Co-Founder @ Roostr — AI-native freight forwarder 2024–now

Outline

I. Intro to Safety and Runtime Assurance

II. Application to Spacecraft Proximity Operations

III. Application to Spacecraft Attitude Control

IV. Planning for Failure

Safety-Critical Control

Given: Input-constrained dynamical system, with **state** $x \in X \subseteq \mathbb{R}^n$ and **input** $u \in U \subset \mathbb{R}^m$

$$\dot{x} = F(x,\ u)$$

Safety is defined *from* constraints on the state:

$$\varphi_i(x) \geq 0, \quad i = 1, \dots, M$$

States satisfying all constraints form the **constraint set** (*allowable set*):

$$\mathcal{C}_A := \{x \in X \mid \varphi_i(x) \geq 0, \quad i = 1, \dots, M\}$$

Example: $\mathcal{C}_A$ specifies "no collisions"

Safety of a Closed Loop System

Safety means satisfying constraints **for all time**.

Forward Invariance

A closed set $S \subset X$ is **forward invariant** w.r.t. a closed-loop system if:

$$x(t_0) \in S \quad \Rightarrow \quad \forall\, t \geq t_0, \quad x(t) \in S$$

Safety: If $u : X \to U$ renders the closed-loop system forward invariant in $\mathcal{C}_S \subseteq \mathcal{C}_A$:

- u is a **safe control law**
- $\mathcal{C}_S$ is the **safe set**
- $x \in \mathcal{C}_S$ is a **safe state**

Control Invariance

So far: given a control law, we can tell if its safe

Control invariance tells you whether its **possible** to find such a safe control law:

Control Invariance

A closed set $S \subset X$ is **control invariant** with respect to a dynamical system and input set U if there exists a control law $u : X \to U$ that renders S forward invariant.

Viability Kernel is the largest control invariant subset.

- A larger safe set means more states are reachable while remaining safe

Backup Slides

Additional detail and formal definitions

Formal Safety Definitions

Definition of Safety

A state $x \in X$ is **safe** with respect to a dynamical system, an admissible control law $u : X \to U$, and a constraint set $\mathcal{C}_A$ if x is in a **forward invariant** subset of $\mathcal{C}_A$. Furthermore, u is said to be a **safe control law** inside that set.

Template Slides

Reference examples for building new slides

Section Title

Brief subtitle or context

Content Slide Example

Key point or framing sentence at the top.

- First point with enough detail to stand alone
- Second point
  - Supporting detail
  - Supporting detail
- Third point

Equation Slide Example

The safety filter solves a quadratic program at each timestep:

Active Safety Filter

$$\min_{u} \quad \|u - u_{\text{des}}\|^2$$

$$\text{s.t.} \quad L_f h(x) + L_g h(x)\, u \geq -\alpha\big(h(x)\big)$$

- $h(x) \geq 0$ defines the safe set
- Minimally invasive: follows u_{des} when safe
- Small QP, solves in real time

Two-Column Slide Example

Left: Text or Math

- Explain the concept
- Key equations or points
- Connection to Archer

Right: Figure

Figure goes here

Full Image Slide Example

Full-width figure or screenshot

Brief caption or takeaway below the image.

Side-by-Side Image Example

Left image (4:3)

Right image (4:3)

Source citation here